

Title: total protein determination

Principle: absorbance is measured photometrical using spectrophotometer.

According to Beer Lambert law the intensity of the color produced is proportion to the concentration of the protein of the sample.

Procedure: first test tubes prepared were labelled as blank, sample and standard

1. The pipette was adjusted to 1NI
2. Biuret reagent was added on each of the test tubes using the pipette
3. The pipette was again adjusted to 20µm
4. Serum protein was added to the second test tube
5. Standard protein was added to the third test tube called standard test tube
6. The second and the third test tube were mixed well
7. The test tubes were taken to incubator for ten minutes at 37⁰ C
8. The test tube contents were added to cuvette, each on different cuvette
9. Spectrophotometer was adjusted to 540nm wavelength
10. The cuvette was added to spectrophotometer at high path length of 1cm
11. The absorbance of each cuvette sample was taken from the spectrophotometer reading
12. The data we got, was used to calculate the concentration of proteins.

Materials used:

1. test tube
2. incubator
3. Pipette
4. Cuvette
5. Micropipette tip
6. Spectrophotometer
7. Test tube rack

Chemical used:

1. biuret reagent
2. Serum protein sample
3. Standard protein
4. Distilled water

Result:

- Measurement for blank tube
Absorbance = 0
Wavelength=540 nm
- Measurement for the test tube containing serum protein and biuret reagent
Absorbance=0.581
Wavelength=540nm
- Measurement for standard test tube
Absorbance=0.395
Wavelength=540nm

Calculation:

$$\text{Concentration of test tube} = \frac{A_t}{A_s} * 6 \text{ g / dl}$$

Where A_t = absorbance of the sample test

A_s = absorbance of the standard

A_b = absorbance of the blank tube

- Subtract the absorbance of the blank from each data
 $A_t = 0.581$ $A_s = 0.395$
 $A_b = 0$ $A_b = 0$
 $A_t - A_b = 0.581$ $A_s - A_b = 0.395$

$$\begin{aligned} \text{Concentration of test tube} &= \frac{0.581}{0.395} * 6 \text{ g / dl} \\ &= 8.825 \text{ g / dl} \end{aligned}$$

Discussion:

Limitations:

- We should adjust the spectrometer to the original point before adding another cuvette sample.
- The cuvette was held on rough side to make the light pass through the smooth side of the cuvette.

- Before the sample is used to be seen by spectrometer, it should be incubated at 37°C, so that the proteins don't denature.

Conclusion:

The conducted experiment showed that the person has ill condition. Therefore, the patient need medical treatment and further diagnosis is needed to explain the case.

References:

Cleveland Clinics. pathology & laboratory medical institute (R. Tomsich)

Principle of spectrometer and it's applications(chemistry). Bjus- Foyers pm, Macmin D. Quality life.